

Zack Freeman

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Education

University of Utah

Expected graduation Fall 2026

Master of Computing: Artificial Intelligence Specialization

University of Utah

May 2025

Bachelor of Science: Computer Science

GPA: 3.90

Bachelor of Science: Physics with an emphasis in Astronomy and Astrophysics

GPA: 3.90

- Relevant Courses: Deep Learning – Machine Learning – Algorithms – Foundations of Data Analysis – Intro to Cosmology – Algorithms and Data Structures – Differential Equations and Linear Algebra – Extragalactic Astrophysics – Databases – Quantum Mechanics

Weber State University

May 2020

Associates of Science

GPA: 3.92

Projects

Sourcerer

- Developed an app that streamlines the process of finding high-quality sources, providing summaries and scores for said sources.
- Built a scalable data pipeline using **Google Cloud Run**, integrated **Dockerized open-source LLMs** with **Ollama**, and trained custom **ML models**.
- Designed and maintained a fully automated **CI/CD pipeline** using **GitLab Runners**, ensuring rapid, reliable deployment.

Topic Classification Model

- Designed a **quantized neural network** trained on **TF-IDF vectors** to classify text into broad categories (e.g., Sports, Politics, etc.) with **85% accuracy**, using **PyTorch** and **scikit-learn**.
- Conducted dataset selection and parameter optimization across various models and feature sets to maximize classification performance, while maintaining small model sizes and fast predictions.

Experience

University of Utah

Teaching Assistant: Foundations of Data Analysis – Dr. El Kindi Rezig

Present

- Assist in assignment development and grading, as well as hold office hours to help students with any problems they may encounter in assignments, quizzes, tests, or any general questions they may have about the material.

Research Assistant – Dr. Anil Seth

May 2022 – Present

- Searching for black hole microlensing events in a Hubble Space Telescope (HST) dataset of ~180,000 stars spanning 20 years of observations. This work includes algorithm development and optimization, parallelization of fitting code, image processing, and ML model development.

Undergraduate Student Advisory Committee (UgSAC) – Kahlert School of Computing (KSoC)

President

Fall 2022 – Fall 2024

- Organized meetings between committee members and faculty, delegated tasks to members, and determined committee goals, in addition to the responsibilities as a member.

Member

Fall 2021 – Spring 2022

- Write formal Retention, Promotion, and Tenure (RPT) reports for incoming and current KSoC faculty, in addition to serving as a liaison between students and faculty.

Learning Assistant: Foundations of Astronomy – Dr. Gail Zasowski

Fall 2022

- Assisted in development and led ~30-person weekly discussions that involved introductory astrophysics topics, such as orbital mechanics and stellar evolution. Nominee for Learning Assistant of the year.

Physics and Astronomy (PANDA) Mentorship Program

Committee Member

Fall 2022

- Designed activities that assisted mentors with teaching their mentees the soft skills needed to navigate the university experience with a team of ~10 professors and students.

Mentor

Spring 2022

- Assisted incoming Physics and Astronomy students with developing soft skills such as communication with professors, time management, etc.

The Front Climbing Club

Risk Management Specialist

May 2021 – May 2022

- Maintained and serviced gym equipment to ensure customer safety. This involved servicing auto-belays, ropes, and all other sport climbing equipment on a regular basis.

Front Desk Staff

May 2020 – May 2022

- Managed front desk operations as Front Desk Lead. Tasks included teaching introductory climbing classes, assisting with events, maintaining the gym, and assisting customers with gear.

Civil Engineering Consultants

Field Inspector | Designer

May 2017 – April 2020

- Performed design in AutoCAD, responsible for field work, and inspected ongoing projects.

Publications and Presentations

Candidate Microlensing Events in Omega Centauri, Freeman, Z., Häberle, M., Seth, A., et al., in preparation

IAU Symposium 398 / MODEST-25 Poster Session	June 2025
Max Planck Institute for Astronomy (MPIA) Presentation	June 2024
Undergraduate Research Symposium	August 2023
Utah Conference on Undergraduate Research	February 2023
American Astronomical Society (AAS)	January 2023
Undergraduate Research Symposium	August 2022

Scholarships and Awards

Dean's List Award	10 semesters \$0
KSoC COMPUTE Magazine Featured Undergraduate Student	Fall 2024 \$0
Learning Assistant of the Year Nominee – Department of Physics and Astronomy	Fall 2022 \$0
Outstanding Continuing Undergraduate Award – Department of Physics and Astronomy	Merit \$1,000
KSoC Departmental Scholarship	Merit \$5,000
Goldman Sachs Scholarship	Merit \$5,000
New Century Scholarship	Merit \$4,000
C.M. Collins Endowed Scholarship	Merit \$3,000
Trustee's Scholarship	Merit \$3,000
Guodong Li Scholarship	Merit \$3,000
David S. Anderson Scholarship	Merit \$2,000
Walter W. Wada Endowed Scholarship	Merit \$1,000
College of Engineering Giving Day Scholarship	Merit \$1,000
Early College Achievement (Weber State University)	Merit \$0
High Honors (Weber State University)	Merit \$0
High Academic Excellence (Weber State University)	Merit \$0

Total Amount Awarded: \$28,000